

## New Sustainability Practices: Renewable Energy



### Renewable Energy

#### Renewable Energy: Meaning

- **Renewable energy** is energy derived from natural sources that are replenished naturally on a human time scale.
- Major sources include:
  - Sunlight
  - Wind
  - Water
  - Biomass
  - geothermal heat
  - and ocean energy.
- Renewable energy is commonly referred to as green or clean energy.
- Renewable resources have the capacity for natural replenishment or self-renewal.

### Renewable Energy

#### Why Renewable Energy?

Conventional energy sources, particularly fossil fuels, have significant environmental consequences.

#### Renewable energy can:

- Reduce dependence on **finite fossil fuels**.
- Reduce harmful **air emissions**.
- Contribute to reducing greenhouse gas emissions.
- Support long-term **energy security**.
- Reduce some forms of waste and environmental pollution.
- Support the transition toward **sustainable development**.

### Renewable Energy

#### Key Characteristics of Renewable Energy:

#### Renewable energy resources are generally:

- Naturally available
- Self-renewing
- Replenishable
- Abundant in the natural environment
- Less dependent on finite resources
- Capable of supporting long-term energy supply
- Generally cleaner than conventional fossil-fuel energy

### Renewable Energy

#### Major Types of Renewable Energy

Seven major forms of renewable energy can be identified:

1. Solar energy
2. Wind energy
3. Hydropower
4. Biomass energy
5. Biofuels
6. Geothermal energy
7. Ocean energy

### Renewable Energy

#### Solar Energy:

**Solar energy** is energy received from the Sun as solar radiation and used for **heating and electricity generation**.

#### Passive Solar Energy

- Uses solar energy through **natural processes** without mechanical equipment.
- Does not directly require electricity for operation.

#### Examples:

- Building orientation and design
- Natural space heating
- Greenhouse heating
- Natural lighting



- Traditionally used in building design for **heating and cooling**.

### Renewable Energy

#### Active Solar Energy:

Uses **technology** to capture and convert solar radiation.

#### Solar Thermal Collectors

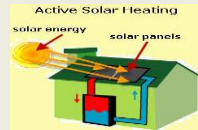
- Convert sunlight into **heat**
- Used for water and space heating
- → Collector → **Heat**

#### Photovoltaic (PV) Cells

- Convert sunlight directly into **electricity**
- Used in rooftop panels and solar farms
- → PV Cells → **Electricity**

**Overall advantages:** Abundant & renewable • Low emissions • Low operating cost • Scalable

**Limitations:** Sunlight-dependent • Space requirement • High initial cost • Storage may be needed



### Renewable Energy

#### Wind Energy:

**Wind energy** converts the kinetic energy of moving air into electricity using wind turbines.

#### Traditional Uses

- Propelling ships
- Operating windmills
- Grinding grain & pumping water

#### Suitable Locations

- Coastal areas
- Open plains
- Mountain areas

**Advantages:** Renewable • No fuel required • Low operational emissions

**Limitations:** Variable wind • Bird impacts • Noise/visual impacts



### Renewable Energy

#### Hydropower:

**Hydropower** converts the energy of **flowing or falling water** into electricity.

#### How it works:

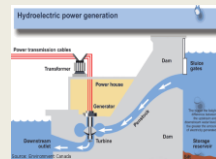
💧 **Water Flow** → Turbine → Generator → Electricity Uses

- Historically: waterwheels for mills and machinery
- Modern: turbines, generators & transformers

**Bangladesh example:** Kaptai Hydroelectric Power Plant

**Advantages:** Renewable • Efficient • Reliable power generation

**Limitations:** High construction cost • Ecosystem alteration • River-flow changes • Social impacts



### Renewable Energy

#### Biomass and Biofuels:

**Biomass energy** is produced from **organic materials** and can be converted into heat, electricity or transportation fuels.

#### Major Sources

- Agricultural & crop residues
- Wood and plant materials
- Animal residues
- Municipal & industrial organic wastes
- Landfill methane

**Uses:** ⚡ Electricity • 🔥 Heating • 🚗 Transportation fuels

**Benefit:** Converts **waste** into **useful energy**.

**Limitations:** Deforestation • Soil degradation • Vegetation degradation



### Renewable Energy

#### Geothermal Energy:

**Geothermal energy** is energy obtained from **heat within the Earth** and used for electricity generation and heating.

#### Uses:

- ⚡ Electricity
- 🔥 Heating
- 🌡 Thermal applications

#### Advantages:

- Reliable • High efficiency • Improves energy security • Lower emissions than fossil fuels

#### Limitations:

- High initial cost • Some GHG emissions • Possible resource overexploitation • Dependent on suitable geological conditions

### Renewable Energy

#### Ocean and Tidal Energy:

**Ocean energy** harnesses energy from **waves, tides, currents, ocean temperature differences, and salinity gradients**.

**Major Sources:** 🌊 Waves • 🌊 Tides • 🌊 Tidal currents • 🌊 Ocean currents • 🔥 Thermal energy • 🌊 Salinity gradients

**Tidal Energy:** Converts the movement of **tides** into **electricity** and is highly predictable.

**Advantages:** Renewable • Abundant • Predictable • Low direct emissions

**Challenges:** High technology cost • Limited suitable locations • Potential marine impacts • Developing technology

### Renewable Energy

#### Major Advantages of Renewable Energy:

Renewable energy can provide **environmental, economic and social** benefits.

##### Environmental

- Lower dependence on fossil fuels
- Reduced air pollution
- Lower greenhouse gas emissions
- Reduced depletion of non-renewable resources

##### Economic

- Diversification of energy sources
- Reduced dependence on imported fuels
- Development of new technologies and industries

##### Social

- Employment opportunities
- Rural economic development
- Improved access to energy
- Greater energy security

### Renewable Energy

#### Challenges of Renewable Energy

Despite its benefits, renewable energy faces several **technical, economic, environmental, and spatial** challenges:

- **Intermittency:** Solar and wind availability varies over time.
- **High initial cost:** Some technologies require substantial investment.
- **Energy storage:** Batteries or other storage may be required.
- **Geographical limitations:** Renewable resources are unevenly distributed.
- **Land-use conflicts:** Large projects may require extensive areas.
- **Environmental impacts:** Hydropower can alter aquatic ecosystems; wind turbines may affect wildlife.
- **Resource overuse:** Unsustainable biomass use can contribute to deforestation.
- **Technology limitations:** Ocean energy remains relatively costly and less technologically developed.

### Renewable Energy

#### Renewable Energy Is Not Always Sustainable:

A renewable resource can **regenerate naturally**, but its use may still cause environmental damage if poorly managed.

##### Example: Biomass

- Biomass is renewable because vegetation can regenerate.
- Excessive harvesting for fuel may cause:
  -  Deforestation
  -  Vegetation degradation
  -  Soil degradation
  -  Loss of ecosystem services
- Heavy dependence on local biomass can lead to **resource overexploitation**.

# Thank you all

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